

Your Paper's Title(SHUMOJIAYOUZHAN)

Summary

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Keywords: keyword1; keyword2

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1 Introduction

1.1 Problem Background

As human civilization expands, Earth's fragile ecosystems face severe pressures from population surges, resource depletion, and environmental degradation, driving a resurgence in space exploration. The Moon, with its proximity advantage and potential as a gateway to deep space, emerges as the foremost candidate for permanent settlement. The space elevator system, a revolutionary infrastructure, utilizes electrically powered graphene cables (with a strength of 130 GPa) to efficiently transport materials from equatorial ground ports to geostationary orbit (GEO) and the top anchor point. Each port has an annual transport capacity of 179,000 metric tons, with three ports synergistically reaching 537,000 metric tons—reducing fuel consumption and pollution by 90%. The Moon Colonization Management (MCM) agency plans to launch a 100,000-person base by 2050, requiring 100 million tons of materials and sustained water supply. Traditional rockets (with an annual transport capacity of only 78,000 tons) would take millennia to accomplish this, while the space elevator could reduce the timeline to decades. However, challenges in cable maintenance and system reliability must be addressed. This paper evaluates pure elevator, pure rocket, and hybrid solutions using Markov failure analysis to assess cost, time, reliability, and environmental impact, providing MCM with optimal logistics decision-making.

1.2 Problem Restatement

The proposed construction of a large-scale lunar colony represents a critical step in humanity's long-term expansion into space, yet it also introduces substantial logistical, economic, and environmental challenges. The status quo of Earth-to-Moon transportation technologies demands a systematic solution to balance construction efficiency, long-term sustainability, and environmental responsibility. Based on a comprehensive analysis of the problem context, the objectives of this study are restated as follows:

- Construct a mathematical model to evaluate the cost and time efficiency of transporting materials from Earth to a lunar colony using different transportation schemes, including space elevator systems, traditional rocket launches, and hybrid approaches. The model should account for large-scale construction demands as well as ongoing supply requirements.
- Incorporate system uncertainties and operational disruptions into the model, such as partial failures of space elevator components or rocket launch inefficiencies, and analyze how these factors affect overall transportation performance, cost, and project timelines.
- Perform a sensitivity analysis to identify the most influential parameters in the transportation system, including payload capacity, transportation rate, failure probability, and environmental impact, and evaluate the robustness of the proposed solutions under varying assumptions.

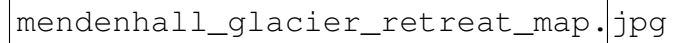
A rectangular box containing the text "mendenhall_glacier_retreat_map.jpg". This box represents a map showing the retreat of Mendenhall Glacier.A rectangular box containing the text "juneau_tourism_concerns_chart.jpg". This box represents a chart showing tourism-related concerns from Juneau residents.

Figure 1: Retreat of Mendenhall Glacier and tourism-related concerns from Juneau residents

- Extend the model to post-construction operations by estimating the annual water demand for a fully operational lunar colony of 100,000 residents and analyzing the additional cost and time required to sustain the population during its first year of habitation.
- Communicate the results through a policy-oriented recommendation, summarizing model outcomes, comparing alternative transportation strategies, and proposing an optimal plan to support the construction and long-term operation of the lunar colony while minimizing adverse environmental effects on Earth.

1.3 Our Work

2 Preparation of the Models

2.1 Assumptions

2.2 Notation

Table 1: Notations and Definitions

Symbol	Definition	Unit	Value/Est.
Decision Variables			
M_{SE}	Total mass transported via Space Elevator system	ton	≥ 0
N_{Rock}	Total number of Falcon Heavy launches	-	Integer
T	Total project duration (Time-to-Completion)	year	≥ 0
Global Parameters			
M_{goal}	Total material mass required for the Moon Colony	ton	10^8
Cost & Capacity Parameters			
C_{launch}	Cost per single Falcon Heavy launch	USD	1.5×10^8
C_{elec_unit}	Electricity cost per ton via Space Elevator	USD/ton	4.15^*
C_{maint}	Annual maintenance cost for SE system	USD/yr	1.2×10^8
C_{TV_fixed}	Fixed cost for Apex-to-Moon Transfer system	USD/yr	3.0×10^8
Rate Constraints			
Cap_{SE}	Annual capacity of the Space Elevator system	ton/yr	5.37×10^5
Cap_{Rock}	Payload capacity of a single rocket launch	ton	125
f_{total}	Max. annual Earth launch frequency (all sites combined)	launches/yr	Data-driven
f_{cycle}	Max. round-trips per rocket per year (apex-to-Moon)	cycles/yr	$\approx 60^\dagger$

*Note: C_{elec_unit} is derived from the weekly energy consumption (E_{week}) and annual elevator throughput.

† Note: f_{cycle} corresponds to a 6-day round trip cycle (approximately 365/6 per year).

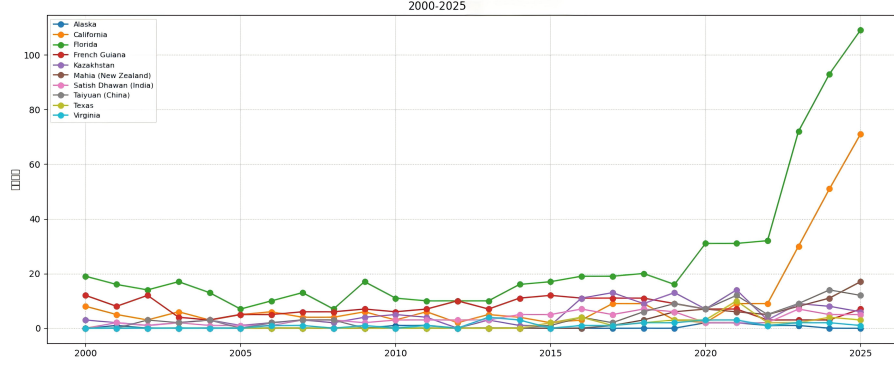


Figure 2: aa

Table 2: Launch-Site-Level Forecast of Annual Launch Frequency in 2050 (Aggressive Extrapolation Scenario)

Launch Site	2025	Annual Growth Rate	2050 Forecast
Florida (USA)	109	13%	≈ 2300 launches/year
California (USA)	71	11%	≈ 950 launches/year
Texas (USA)	3	18%	≈ 190 launches/year
Virginia (USA)	1	16%	≈ 70 launches/year
Mahia (New Zealand)	17	9%	≈ 140 launches/year
Taiyuan (China)	12	9%	≈ 100 launches/year
Satish Dhawan (India)	5	8%	≈ 35 launches/year
Kazakhstan (Baikonur)	6	6%	≈ 25 launches/year
French Guiana (Kourou)	7	5%	≈ 24 launches/year

Note: This table presents an upper-bound forecast based on exponential extrapolation of average growth rates observed during 2020–2025. No constraints related to launch-site capacity, regulatory limitations, or market saturation are considered. Results should therefore be interpreted as a high-growth scenario rather than a deterministic prediction.

2.3 Data Preparation

To model the long-term reduction in launch cost, an experience-curve-based exponential decay model is adopted. In the aerospace industry, a well-established empirical rule suggests that unit cost decreases by approximately 15–20% with each doubling of cumulative production. This learning effect is captured by the following formulation:

$$Cost(t) = C_0 \exp[-k(t - t_0)], \quad (1)$$

where C_0 denotes the baseline expendable launch cost and k represents the learning rate associated with increased production scale and operational maturity. Using a reference cost of $C_0 = \$150$ million and calibrating the model over a 24-year horizon, the parameter k is selected as $k \approx 0.096$, which lies within the commonly accepted range $0.05 \leq k \leq 0.10$. Under this calibration, the model yields an average single-launch cost of approximately \$15 million, corresponding to an order-of-magnitude reduction relative

to the expendable baseline. This value is consistent with an aggressive scenario characterized by full reusability, high launch cadence, and mature operational efficiency, and is therefore adopted as the model-implied marginal launch cost in subsequent analyses.

Table 3: Falcon Heavy–Class Launch Cost Model Inputs (Assumption Set)

Parameter	Real-World Reference	Model Input	
Single-launch cost	~ \$150M (expendable baseline)	\$15M (average)	Assumes fully mature
Payload per launch	~20 t (TLI reference)	125 t (average)	Strictly follows
Reuse lifetime	< 20 flights	100 flights	Enabled by materials/
Propellant type	RP-1 (kerosene)	Methalox (CH₄/LOX)	Lower coking ⇒ I

3 Model

3.1 Model 1

Scenario A: Space Elevator–Dominated Transportation System (Pure Elevator Scenario)

In this scenario, all construction materials required for the lunar settlement are first transported from the Earth’s surface to the orbital anchor point via the Galactic Harbour space elevator system. The elevator provides a continuous and pollution-free logistics channel from Earth to orbit, operating at a fixed annual throughput determined by its design capacity.

After the initial deployment phase, transport rockets are permanently stationed in orbit and no longer return to Earth. Each rocket is initially launched from the Earth’s surface without payload and transferred to the orbital anchor point, where it enters a repetitive operational cycle. In each cycle, a rocket loads materials delivered by the space elevator at the anchor point, transports them to the lunar settlement, unloads the payload, and then returns to the anchor point for the next loading operation. All subsequent material transfers between Earth and the Moon are therefore completed through the combined operation of the space elevator (Earth to anchor point) and the orbital rocket fleet (anchor point to Moon).

Under this scenario, rockets are used exclusively for cislunar transportation and do not perform repeated Earth surface launches. As a result, the transportation system avoids the high fuel consumption and environmental impact associated with frequent atmospheric launches. The overall delivery rate to the Moon is jointly constrained by the elevator’s upward transport capacity and the total carrying capacity of the orbital rocket fleet.

Scenario B: Conventional Rocket-Based Transportation System (Pure Rocket Scenario)

In the pure rocket scenario, all materials required for lunar base construction are transported directly from the Earth’s surface to the Moon using conventional launch vehicles. Each transportation cycle consists of a fully loaded rocket launch from a ground-

based launch site, insertion into a translunar trajectory, delivery of payload to the lunar settlement, and a return flight to the original launch site.

Rockets repeatedly execute the complete launch–transport–return cycle throughout the project duration until all required materials have been delivered. No orbital infrastructure, such as a space elevator or an anchor-point logistics hub, is utilized in this scenario. Consequently, every transportation task requires overcoming Earth’s gravitational well from the surface, leading to substantial fuel consumption and high operational costs per delivery.

While this approach benefits from a relatively simple system architecture and does not require the construction or maintenance of orbital facilities, its overall transportation efficiency is limited by launch frequency, fuel requirements, and the cumulative cost of repeated Earth-based launches. The total delivery rate to the lunar settlement is determined by the annual number of launches and the payload capacity of each rocket.

Scenario C: Hybrid Space Elevator–Rocket Transportation System (Mixed Scenario)

The hybrid scenario integrates the advantages of both the space elevator and conventional rockets by dividing the transportation process into two operational phases. In the initial phase, rockets are launched from the Earth’s surface with full payloads and deliver materials directly to the lunar settlement. After completing this first delivery, instead of returning to Earth, these rockets are transferred to the orbital anchor point and retained in orbit as part of a permanent transportation fleet.

In the second phase, the retained rockets operate in a closed-loop cislunar logistics cycle. Materials are continuously transported from the Earth’s surface to the orbital anchor point by the space elevator, where they are loaded onto the orbiting rockets. Each rocket then transports the payload from the anchor point to the Moon and returns to the anchor point empty, ready for the next cycle. This process is repeated until all remaining materials required for lunar base construction have been delivered.

By allowing rockets to perform a single Earth-based launch before transitioning to orbital operations, the hybrid scenario reduces the number of costly and fuel-intensive surface launches while maintaining a flexible and scalable transportation capacity. The system’s performance depends on the balance between the space elevator’s supply rate and the number of rockets retained in orbit, enabling a trade-off between total transportation time and overall launch cost.

The following are definitions for various status:

- (1) Launch the empty rocket from the ground to the top anchor point of Galaxy Port, or transport rocket components via the Galaxy Port elevator to the top anchor point for assembly;
- (2) Launch the fully loaded rocket from the ground into the Earth-Moon transfer orbit;
- (3) Depart the fully loaded rocket from the Earth-Moon transfer orbit to the lunar colony for unloading;

- (4) Empty rocket is retrieved from the lunar colony and returned to the top anchor point of Galaxy Port;
- (5) Empty rocket is retrieved from the Earth-Moon transfer orbit and returned to its original launch site.

Based on the above fundamental procedures, the three transport scenarios can be formally described from the operational perspective of a single rocket:

3.1.1 Mass Requirement

The total mass required for constructing the lunar colony is

$$M_{goal} = 10^8 \text{ ton.} \quad (2)$$

Any feasible transportation plan must satisfy the total delivered mass constraint:

$$M_{delivered} \geq M_{goal}. \quad (3)$$

3.1.2 Space Elevator Capacity Constraint

The total mass transported by the Space Elevator system is bounded by its annual capacity:

$$M_{SE} \leq Cap_{SE} \cdot T. \quad (4)$$

3.1.3 Rocket Launch Frequency Constraint (Earth-based)

Given the maximum annual launch frequency across all Earth launch sites,

$$N_{Rock} \leq f_{total} \cdot T. \quad (5)$$

3.1.4 Orbital Rocket Fleet Throughput (Apex-to-Moon Cycling)

Assuming each launched rocket can be retained to operate in orbit and perform at most f_{cycle} round trips per year (Apex $\leftrightarrow$ Moon), the upper bound of Apex-to-Moon delivered mass is

$$M_{Apex \rightarrow Moon} \leq N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T. \quad (6)$$

3.1.5 Scenario-Specific Delivered Mass

We formalize the three scenarios via different expressions of $M_{delivered}$:

Scenario A: Pure Elevator Scenario (Elevator-dominated) All cargo first reaches the orbital anchor point via the Space Elevator, thus the Moon delivery is limited by both the elevator supply and orbital cycling capability:

$$M_{delivered}^{(A)} \leq \min \left(M_{SE}, N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T \right). \quad (7)$$

The feasibility requirement becomes

$$\min \left(M_{SE}, N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T \right) \geq M_{goal}. \quad (8)$$

Scenario B: Pure Rocket Scenario (Earth-to-Moon direct launches) All cargo is delivered by Earth-based launches. Each launch delivers at most Cap_{Rock} , therefore

$$M_{\text{delivered}}^{(B)} \leq N_{Rock} \cdot Cap_{Rock}. \quad (9)$$

The feasibility requirement becomes

$$N_{Rock} \cdot Cap_{Rock} \geq M_{goal}. \quad (10)$$

Scenario C: Hybrid Scenario (One-time direct + orbital cycling with elevator supply) In the hybrid design, each rocket performs one Earth-to-Moon delivery at deployment, and then participates in the Apex-to-Moon cycling supported by elevator supply. Hence, the total delivered mass can be bounded by

$$M_{\text{delivered}}^{(C)} \leq N_{Rock} \cdot Cap_{Rock} + \min \left(M_{SE}, N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T \right). \quad (11)$$

The feasibility requirement becomes

$$N_{Rock} \cdot Cap_{Rock} + \min \left(M_{SE}, N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T \right) \geq M_{goal}. \quad (12)$$

3.1.6 Cost Model (for Optimization or Result Reporting)

The total cost consists of launch cost, elevator electricity cost, annual maintenance, and a fixed transfer-system cost:

$$C_{\text{total}} = C_{\text{launch}} \cdot N_{Rock} + C_{\text{elec_unit}} \cdot M_{SE} + C_{\text{maint}} \cdot T + C_{TV_fixed}. \quad (13)$$

3.2 Closed-Form Solutions for Scenario A and Scenario B

Scenarios A and B admit direct analytical solutions because their logistics chains contain no internal trade-off once feasibility is enforced. In this section, we derive the minimum completion time and the corresponding resource requirements under each scenario.

3.2.1 Scenario A: Pure Space Elevator Scenario

In Scenario A, all payload must be transported from the Earth surface to the orbital anchor point by the Space Elevator system, and then delivered to the Moon by the orbital rocket fleet. Therefore, the elevator-delivered mass must cover the entire requirement:

$$M_{SE} = M_{goal}. \quad (14)$$

The Space Elevator capacity constraint implies

$$M_{SE} \leq Cap_{SE} \cdot T, \quad (15)$$

which yields the minimum feasible completion time

$$T_A^* = \frac{M_{goal}}{Cap_{SE}}. \quad (16)$$

To prevent long-term accumulation of cargo at the anchor point, the orbital rocket fleet must be able to transport at least as much as the elevator supplies over the project horizon. Using the rocket cycling limit f_{cycle} and per-trip payload Cap_{Rock} , the cumulative Apex-to-Moon transport capacity is

$$M_{Apex \rightarrow Moon} \leq N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T. \quad (17)$$

Feasibility requires $M_{Apex \rightarrow Moon} \geq M_{SE}$, hence

$$N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T_A^* \geq M_{goal}. \quad (18)$$

Substituting $T_A^* = M_{goal}/Cap_{SE}$, the required number of rockets is bounded by

$$N_{Rock} \geq \frac{Cap_{SE}}{f_{cycle} \cdot Cap_{Rock}}, \quad (19)$$

and thus the minimal integer solution is

$$N_{Rock,A}^* = \left\lceil \frac{Cap_{SE}}{f_{cycle} \cdot Cap_{Rock}} \right\rceil. \quad (20)$$

Finally, the total cost under Scenario A can be evaluated by the unified cost model:

$$C_A = C_{launch} \cdot N_{Rock,A}^* + C_{elec_unit} \cdot M_{SE} + C_{maint} \cdot T_A^* + C_{TV_fixed}, \quad (21)$$

where $M_{SE} = M_{goal}$ and $T_A^* = M_{goal}/Cap_{SE}$.

3.2.2 Scenario B: Pure Rocket Scenario

In Scenario B, all payload is delivered directly from the Earth surface to the Moon by conventional rocket launches. Therefore, the total delivered mass is bounded by

$$M_{delivered}^{(B)} \leq N_{Rock} \cdot Cap_{Rock}. \quad (22)$$

Feasibility requires

$$N_{Rock} \cdot Cap_{Rock} \geq M_{goal}, \quad (23)$$

which gives the minimum number of launches

$$N_{Rock,B}^* = \left\lceil \frac{M_{goal}}{Cap_{Rock}} \right\rceil. \quad (24)$$

Given the maximum annual launch frequency f_{total} (all sites combined), the launch-count constraint is

$$N_{Rock} \leq f_{total} \cdot T. \quad (25)$$

Hence, the minimum feasible completion time is obtained by saturating the launch rate:

$$T_B^* = \frac{N_{Rock,B}^*}{f_{total}}. \quad (26)$$

The corresponding total cost under Scenario B is

$$C_B = C_{launch} \cdot N_{Rock,B}^* + C_{maint} \cdot T_B^*. \quad (27)$$

Note that Scenario B does not require Space Elevator electricity costs or the fixed Apex-to-Moon transfer infrastructure cost because no orbital logistics hub is utilized.

3.3 Optimization of the Hybrid Transportation Scenario

Unlike Scenarios A and B, which admit direct analytical solutions, Scenario C (the Hybrid Space Elevator–Rocket Transportation System) involves an inherent trade-off between transportation time and total system cost. Increasing the number of deployed rockets accelerates material delivery but incurs higher launch and operational costs. Therefore, Scenario C is formulated as a bi-objective optimization problem.

3.3.1 Decision Variable

The key decision variable in Scenario C is the total number of rockets launched from Earth and subsequently retained for orbital cycling:

$$N_{Rock} \in \mathbb{Z}_{\geq 0}. \quad (28)$$

Each rocket performs one initial fully loaded Earth-to-Moon delivery during the deployment phase and then joins the Apex-to-Moon cyclic transportation supported by the Space Elevator system.

3.3.2 Delivered Mass Decomposition

The total mass delivered directly to the Moon during the deployment phase is

$$M_{\text{direct}} = N_{Rock} \cdot Cap_{Rock}. \quad (29)$$

The remaining mass must be supplied via the Space Elevator system:

$$M_{SE} = \max(0, M_{\text{goal}} - N_{Rock} \cdot Cap_{Rock}). \quad (30)$$

3.3.3 Effective Transportation Rate and Project Duration

After deployment, the orbital rocket fleet provides an annual Apex-to-Moon transportation capacity of

$$R_{\text{rocket}} = N_{Rock} \cdot f_{\text{cycle}} \cdot Cap_{Rock}. \quad (31)$$

The Space Elevator system provides an annual supply capacity of Cap_{SE} . Hence, the effective annual delivery rate during the steady-state phase is

$$R(N_{Rock}) = \min(Cap_{SE}, N_{Rock} \cdot f_{\text{cycle}} \cdot Cap_{Rock}). \quad (32)$$

The time required to complete delivery of the remaining mass is approximated by

$$T_{\text{remain}}(N_{Rock}) = \frac{M_{SE}}{R(N_{Rock})}. \quad (33)$$

In addition, deploying N_{Rock} rockets from Earth is limited by the maximum annual launch frequency f_{total} , resulting in a deployment time of

$$T_{\text{deploy}}(N_{Rock}) = \frac{N_{Rock}}{f_{total}}. \quad (34)$$

The total project duration is approximated as

$$T(N_{Rock}) = \max(T_{\text{deploy}}(N_{Rock}), T_{\text{remain}}(N_{Rock})). \quad (35)$$

—

3.3.4 Cost Model

The total system cost for Scenario C consists of rocket launch costs, Space Elevator electricity costs, annual maintenance costs, and a fixed transfer system cost:

$$C(N_{Rock}) = C_{\text{launch}} \cdot N_{Rock} + C_{\text{elec_unit}} \cdot M_{SE} + C_{\text{maint}} \cdot T(N_{Rock}) + C_{TV_fixed}. \quad (36)$$

—

3.3.5 Weighted-Sum Multi-Objective Optimization

Scenario C is formulated as a bi-objective minimization problem with objectives of minimizing total project duration and total cost:

$$\min_{N_{Rock}} (T(N_{Rock}), C(N_{Rock})). \quad (37)$$

To obtain Pareto-optimal solutions, the weighted-sum method is adopted. The normalized scalar objective function is defined as

$$J(N_{Rock}) = \alpha \frac{T(N_{Rock})}{T_{\text{ref}}} + (1 - \alpha) \frac{C(N_{Rock})}{C_{\text{ref}}}, \quad \alpha \in [0, 1], \quad (38)$$

where T_{ref} and C_{ref} are reference values used for normalization, chosen as the maximum observed project duration and cost among all feasible solutions.

By varying the weight parameter α , a set of Pareto-optimal solutions is generated, representing different trade-offs between time efficiency and economic cost. The final recommended solution may be selected based on practical preferences, such as a balanced compromise near the knee point of the Pareto frontier.

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3.3.6 Feasible Range of the Decision Variable

To ensure meaningful system operation, the search range of N_{Rock} is bounded as follows:

$$\left\lceil \frac{Cap_{SE}}{f_{cycle} \cdot Cap_{Rock}} \right\rceil \leq N_{Rock} \leq \left\lceil \frac{M_{goal}}{Cap_{Rock}} \right\rceil. \quad (39)$$

The lower bound guarantees that the orbital rocket fleet is capable of matching the Space Elevator supply rate, while the upper bound corresponds to delivering all required mass through direct Earth-to-Moon launches.

3.4 Model 2

4 Validating the Model

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5 Conclusions

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6 A Summary

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7 Evaluate of the Mode

8 Strengths and weaknesses

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8.1 Strengths

- **Applies widely**

This system can be used for many types of airplanes^[1,2], and it also solves the interference during the procedure of the boarding airplane, as described above we can get to the optimization boarding time. We also know that all the service is automate.

- **Improve the quality of the airport service**

Balancing the cost of the cost and the benefit, it will bring in more convenient for airport and passengers. It also saves many human resources for the airline.

-

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- [2] Lamport, Leslie, L^AT_EX: “ A Document Preparation System ”, Addison-Wesley Publishing Company, 1986.
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- [4] <http://www.chinatex.org/>

Appendices

Appendix A First appendix

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue

nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

Here are simulation programmes we used in our model as follow.

Input matlab source:

```
function [t,seat,aisle]=OI6Sim(n,target,seated)
pab=rand(1,n);
for i=1:n
    if pab(i)<0.4
        aisleTime(i)=0;
    else
        aisleTime(i)=trirnd(3.2,7.1,38.7);
    end
end
```

Appendix B Second appendix

some more text **Input C++ source:**

```
//=====
// Name       : Sudoku.cpp
// Version    : a.0
// Copyright   : Your copyright notice
// Description : Sudoku in C++.
//=====

#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int table[9][9];

int main() {

    for(int i = 0; i < 9; i++){
        table[0][i] = i + 1;
    }

    srand((unsigned int)time(NULL));

    shuffle((int *)&table[0], 9);

    while(!put_line(1))
    {
        shuffle((int *)&table[0], 9);
    }

    for(int x = 0; x < 9; x++){
        for(int y = 0; y < 9; y++){
```

```
        cout << table[x][y] << " ";  
    }  
  
    cout << endl;  
}  
cout << "Python is a best language";  
  
return 0;  
}
```

Report on Use of AI

OpenAI-GPT5-Thinking

Query1 : prompt1

Output1:

AI Output